

Guillaume Payeur

PhD Student in Physics

Université de Montréal — Ciela Institute — Mila

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▷ Research

I'm a PhD student in physics at the Université de Montréal, under the supervision of Laurence Perreault-Levasseur. I'm also affiliated with Mila, Ciela and the Trottier Space Institute.

▷ Education

PhD in Physics <i>Université de Montréal</i>	2025 – Present
MSc in Physics (High Energy Theory) <i>McGill University</i>	2023 – 2025 GPA: 4.0/4.0
BSc in Mathematics & Physics (Honours) <i>McGill University</i>	2019 – 2023 GPA: 3.98/4.0
DCS in Pure and Applied Science <i>Champlain College</i>	2017 – 2019

▷ Publications

- Guillaume Payeur, Evan McDonough, and Robert Brandenberger. Do observations prefer thawing quintessence? *Physical Review D*, 111(12):123541, June 2025
<https://journals.aps.org/prd/abstract/10.1103/bggr-61nr>
- Guillaume Payeur, Evan McDonough, and Robert Brandenberger. Swampland conjectures constraints on dark energy from a highly curved field space. *Physical Review D*, 110(10):106011, November 2024
<https://journals.aps.org/prd/abstract/10.1103/PhysRevD.110.106011>
- Edward W. Kolb, Andrew J. Long, Evan McDonough, and Guillaume Payeur. Completely dark matter from rapid-turn multifield inflation. *Journal of High Energy Physics*, 2023(2):181, February 2023
[https://link.springer.com/article/10.1007/JHEP02\(2023\)181](https://link.springer.com/article/10.1007/JHEP02(2023)181)
- Guillaume Payeur, Étienne Artigau, Laurence Perreault Levasseur, and René Doyon. Correlated read noise reduction in infrared arrays using deep learning. *The Astronomical Journal*, 163(6):292, may 2022
<https://iopscience.iop.org/article/10.3847/1538-3881/ac69d2>

▷ Talks

Do Observations Prefer Thawing Quintessence? <i>Dark Side of the Universe</i>	Montreal, 2025
Do Observations Prefer Thawing Quintessence <i>McGill HEP Theory Journal Club</i>	Montreal, 2025
Completely Dark Matter & the de Sitter Swampland Conjecture <i>WITP Summer Symposium</i>	Winnipeg, 2022
Readout Noise & Machine Learning <i>IREX End-of-Summer Presentations</i>	Montreal, 2020

▷ Awards & Funding

NSERC Canada Graduate Scholarship – Doctoral \$120,000	Fall 2025 – Summer 2027
Dr & Mrs Milton Leong Fellowship \$5,000	Fall 2024 – Winter 2025
FRQNT Master’s Training Scholarship \$40,000	Fall 2023 – Winter 2025
NSERC Canada Graduate Scholarship – Master’s \$17,500	Fall 2023 – Winter 2024
Lorne Trottier Fellowship \$5,000	Fall 2023 – Winter 2024
Trottier Space Institute Graduate Fellowship \$2,000	Fall 2023 – Winter 2025
First Class Honours in Mathematics and Physics	Winter 2023
Wing Hing Chang Scholarship in Science \$800	Fall 2021 – Winter 2022
NSERC Undergraduate Student Research Award \$6,000	Summer 2021
McGill Dean’s Honour List	Fall 2020 – Winter 2021
Bourse Hors UdeM \$4,000	Summer 2020
Faculty of Science Scholarship \$700	Fall 2020 – Winter 2021
McGill Dean’s Honour List	Fall 2019 – Winter 2020
Champlain College Dean’s List	Winter 2018
Champlain College Dean’s List	Fall 2018

▷ Teaching

Course Teaching Assistant at McGill

- PHYS 362 Statistical Mechanics (Winter 2025) — 1 lecture
- PHYS 346 Major Quantum Mechanics (Fall 2024) — 1 lecture, 6 tutorials
- PHYS 142 Electromagnetism and Optics (Winter 2024) — 4 tutorials
- PHYS 251 Honours Classical Mechanics I (Fall 2023) — 1 lecture, 4 tutorials

▷ **Other Honors**

Edith Engelberg Teaching Award Finalist
McGill University

2024 – 2025

Second Place, Astromatic Hackathon
Montreal

2022